

Blockchain-based traceability for authenticating sustainability claims in fermented milk products.

Life sciences × Product traceability and digital product passport × Blockchain and DLT · OS068 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
44 is the world moving	16 can we play, can we win	LOW 0 named in the evidence	€321m modelled evidence · per year	LATER research and standards activi...	L3 10 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Life sciences, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity is a blockchain-based traceability system for authenticating sustainability claims in fermented milk products, targeting dairy manufacturers and retailers. It exists now because certification and labelling alone are being questioned, and blockchain research shows promise for transparent, tamper-proof supply chains.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€387m €78.3m – €1.5bn	€321m €65.1m – €1.3bn	€402k €81k – €1.6m	modelled

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

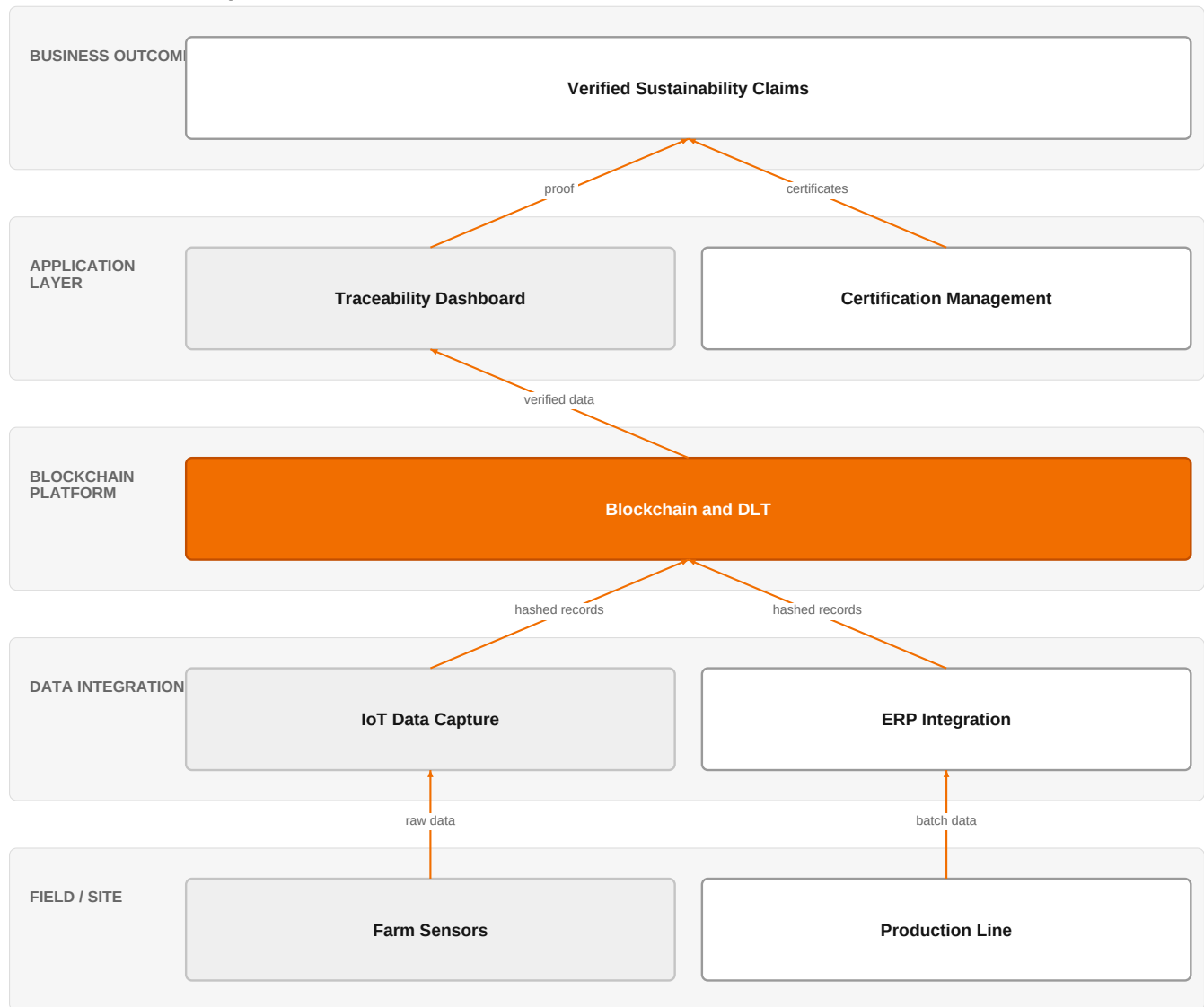
Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Life sciences (EU27_2020)	6,493 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Blockchain and DLT	77.6 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC	2025
Annual value of one engagement	€320k per enterprise per year	assumption	config/sizing.yaml · fallback_bands_eur	size-2026-08-a
Size-weighted buyer base	1,558 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.1 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC is used as a proxy for Blockchain and DLT: No DLT series at NACE level. Contract value is a configured assumption, not observed. Only 0 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

The solution, in one picture

Blockchain Traceability for Fermented Milk



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how IoT data from farms and production lines is anchored on a blockchain to provide tamper-proof evidence for sustainability claims.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Consumer demand for sustainably produced food has intensified, but the authenticity of sustainability claims in fermented milk products remains contested, raising concerns about consumer trust and regulatory clarity. Research is exploring how certification and labelling can verify these claims, while blockchain studies demonstrate its potential to enhance transparency, traceability, and trust in supply chains by addressing data manipulation, fragmented information, and third-party dependencies.

Sources: SIG-A61B4DE2FFB7 [openalex.org 2025-09-19](https://openalex.org/2025-09-19); SIG-EC14886FDE80 [openalex.org 2026-03-22](https://openalex.org/2026-03-22); SIG-1C2398920027 [openalex.org 2026-02-04](https://openalex.org/2026-02-04); SIG-D9CA278082C5 [openalex.org 2025-11-27](https://openalex.org/2025-11-27)

Who buys, and why now

The chief sustainability officer or head of quality at a dairy manufacturer would sign, while the supply chain manager feels the pain of proving origin and handling contamination recalls. They are triggered by a regulator questioning a claim, a retailer demanding proof, or a contamination event that exposes traceability gaps.

Why it is hot now — every claim bound to a dated source

- Certification and labelling are being examined to verify sustainability claims in fermented milk products. [SIG-A61B4DE2FFB7]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts to capture data at production and logistics stages, Healthcare experts to ensure compliance with health-sector standards, and Orange Group researchers to design the blockchain architecture. The engagement would start with a pilot on a specific product line, integrating existing customer systems, and then scale across the supply chain.

Why Orange can win

Orange brings a unique combination of IoT expertise, healthcare domain knowledge, and research capability, plus HDS certification that assures data handling in sensitive environments. However, the asset list is thin for this specific use case, lacking a named blockchain product or reference in dairy, so the win would depend on leveraging the broader Orange ecosystem.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Blockchain and DLT	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
HDS (Hebergeur de Donnees de Sante)	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
MicroPort CardioFlow	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

LOW competitive intensity (score 1.2; 0 of 8 listed players appear in this topic's own evidence)

One or two adjacent players. Orange would be shaping the space rather than fighting for it.

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	competes in OX: Smart Industries — also an Orange partner	structural	—
Accenture	Global systems integrator	competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	competes in OX: Smart Industries	structural	—
NTT DATA	Global systems integrator	competes in OX: Smart Industries	structural	—
Telia Company	Telecom operator peer	competes in OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	competes in OX: Smart Industries	structural	—
Actility	Industrial / OT specialist	competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How do you currently verify the sustainability claims on your fermented milk products, and what triggered you to look for a better method?

- 2 Which specific claims (e.g., organic, carbon-neutral, animal welfare) are hardest to prove today, and to whom do you need to prove them?
- 3 Have you experienced a recall or a customer dispute where traceability gaps were exposed? What happened?
- 4 What systems do you use today for tracking raw milk from farm to factory, and where are the handoffs that lose data?
- 5 Who in your organisation would own a traceability initiative—quality, IT, or sustainability—and how would they measure success?
- 6 Are you under any current or anticipated regulatory pressure to provide more transparent supply chain data?

Objections you will meet

They will say	You can answer
We already have a traceability system in place; why do we need blockchain?	That's fair—many companies have basic traceability. But the issue is trust: your current system may be centralized and vulnerable to data manipulation, as research shows. Blockchain adds immutability and transparency that can strengthen the credibility of your sustainability claims.
This sounds expensive and complex to integrate with our existing IT.	You're right that integration is a real consideration. That's why we'd start with a pilot on a single product line, using our IoT experts to connect to your existing systems, and scale only after proving value. We can also leverage our HDS certification to ensure data security.
We're not sure the regulatory environment will require this level of traceability.	That's a valid concern. However, consumer trust is already at stake, and research shows that certification and labelling are being scrutinized. By acting now, you can get ahead of potential regulation and differentiate your brand.

Risks and unknowns

The main risk is that the customer sees this as a nice-to-have rather than a regulatory necessity, stalling the sale. Unknowns include the specific regulatory trajectory for sustainability claims in dairy, and whether the customer's existing systems can integrate with a blockchain layer without significant rework.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European dairy cooperative that differentiates Orange's offer by leveraging the HDS certification and the MicroPort CardioFlow reference to demonstrate our ability to handle sensitive data and deliver trusted traceability solutions.
Sales	In a conversation with a European dairy cooperative's head of quality, say: 'We have deep expertise in IoT and healthcare data security, and we're exploring how blockchain can make sustainability claims in fermented milk verifiable end-to-end — could we walk through your current certification and labelling process together?'
Strategist / Innovator	Commission a deep-dive brief on how Orange's IoT and healthcare expertise could underpin a blockchain-based traceability service for verifying sustainability claims in fermented milk products, focusing on the certification and labelling angle.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-3C78D9B435E4	2026-07-27	Coffee Traceability Demon...	1	Coffee Traceability Demonstrator Based on Digital Product Passport Principles
SIG-EC14886FDE80	2026-03-22	openalex.org	1	A Blockchain-Enabled Smart Contract Architecture for Enhancing Transparency, Traceability, and ...
SIG-85F36B96E175	2026-03-01	openalex.org	1	The Integration of RFID and Blockchain Technology for the Supply Chain Traceability of Durians
SIG-A85B5F017CD1	2026-02-27	openalex.org	1	Blockchain and Big Data Integration for Transparent and Traceable Supply Chains
SIG-1C2398920027	2026-02-04	openalex.org	1	Enhancing fruit supply chain traceability through blockchain and cryptographic protocols for ac...
SIG-D9CA278082C5	2025-11-27	openalex.org	1	A Blockchain-Based Approach to Enhancing Traceability and Transparency in the Supply Chain
SIG-A0D6903DEEC3	2025-11-19	openalex.org	1	Design of a Blockchain-Enabled Traceability System for Pleurotus ostreatus Supply Chains
SIG-14060280DE3E	2025-11-06	openalex.org	1	Enhancing Agricultural Supply Chain Traceability with Blockchain, Smart Contracts, and E-Label...
SIG-A61B4DE2FFB7	2025-09-19	openalex.org	1	Role of Certifications and Labelling in Ensuring Authenticity and Sustainability of Fermented M...
SIG-25647612A208	2025-08-30	openalex.org	1	A Narrative Review and Gap Analysis of Blockchain for Transparency, Traceability, and Trust in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:16+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:19+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:56:41+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.